

Introduction

Smart Parking system is the first step towards understanding how our vehicle works and how amazingly it has embedded sensors. In real cars em waves are used to sense the obstacle and radar is placed in the car bumper to catch it. We implemented this project by using ultrasonic sensor and Arduino UNO to give it a start.

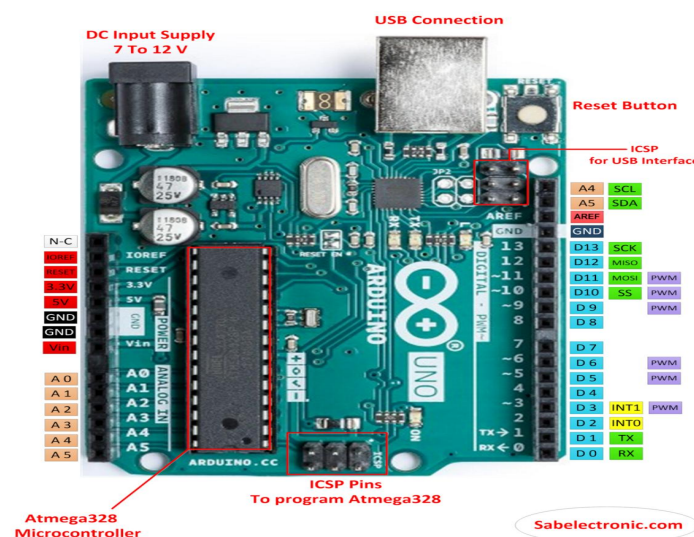
Components Used

1. Arduino Uno
2. Ping Ultrasonic Sensor
3. DC supply
4. Piezo Buzzer
5. Traffic Light Led's
6. Potentiometer
7. LCD Display

Arduino UNO

It is a single board computer featuring AtMega 328P microcontroller. Arduino is a company who manufacture these. It is a company from Italy which makes boards and has a family of micro controller boards like UNO, Mega, Nano etc. It consists of GPIO Pins which are further divided into analog and digital. Total 14 Pins.

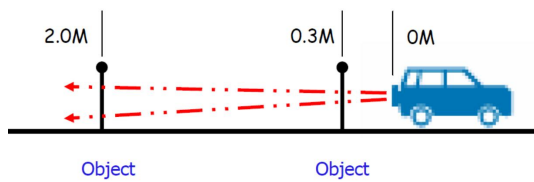
Digital pin work is to take high and low while analog take more than 2 values. It has built in ADC and DAC.



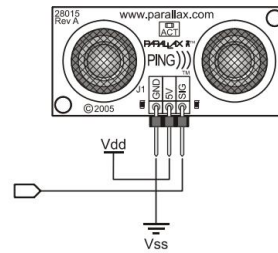
Arduino Uno Diagram Source adapted from [1]

Ping Ultrasonic Sensor

It is a ultrasonic sensor used to send ultrasonic burst and listen for echo at the SIG pin.It is expensive although highly accurate.It is not commonly used.



Source adapted from [2]



PING Module source adapted from [3]

Other Components:

Serial LCD display is used because it is more faster than i2c. The LCD display connects to the Arduino using a series of pins. The specific connections depend on the LCD model and how you want to control it (4-bit vs. 8-bit mode). Generally, you'll connect the LCD's RS (Register Select), EN (Enable), and data pins (D4-D7) to Arduino's digital pins. You might also need a contrast control pin and backlight control (if desired). A piezo buzzer has a piezoelectric ceramic disc inside. When a voltage is applied across this disc, it vibrates and produces sound waves. You can control the pitch (frequency) and duration of the sound by manipulating the electrical signal sent to the buzzer using your Arduino code.



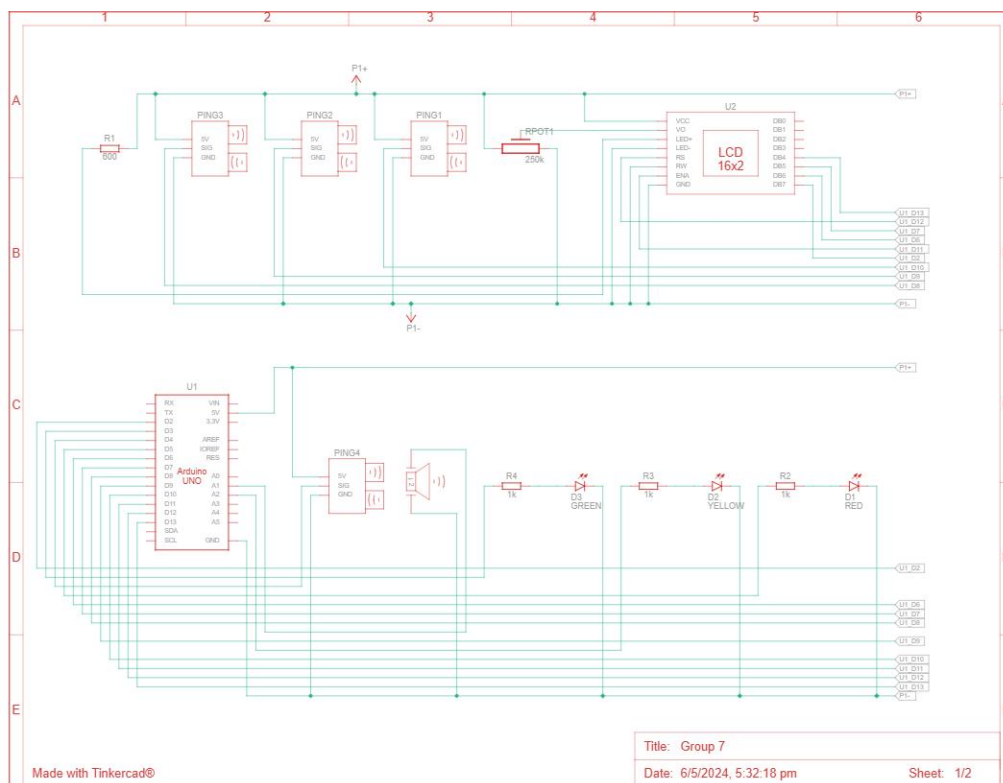
LED Pin out adapted from [5]

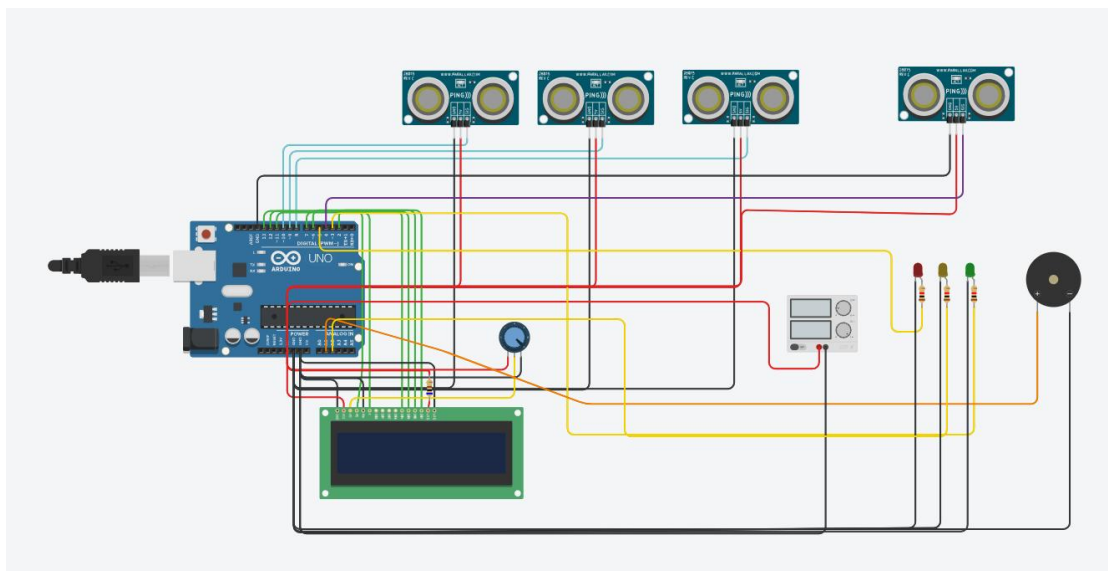
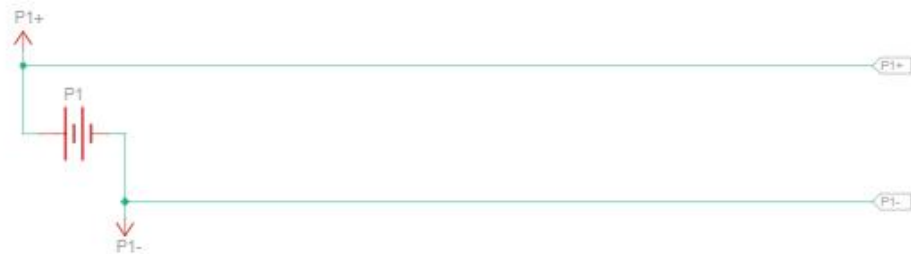
Ps1240 source adapted from [4]



Circuit Diagram

In this circuit diagram there is 4 ultrasonic sensor is used. 1-3 sensors are used for slot detection weather the vehicle are present or not. The 4th sensor is used for vehicle to park in free slots. The 1-3 sensors check weather the vehicle is parked or not by giving an value if sensor detects within the range some vehicle may present that slot is full. Same as the 4 sensor used for parking vehicle without demolition weather the vehicle is comes within closest range the red like will be shown and buzzer gets on. Weather the vehicle in second range the yellow light on same as weather the vehicle in third range green light will be on.





Made by using Tinker cad tool

Arduino Sketch Code

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 13, 7, 6, 2);

#define t1 10
#define t2 9
#define t3 8
#define t4 4
#define Green 3
#define Yellow A2
#define Red 5
```

```

#define buzzer A1
int distanceThreshold = 100;

void setup() {
  pinMode(Green, OUTPUT);
  pinMode(Yellow, OUTPUT);
  pinMode(Red, OUTPUT);
  pinMode(buzzer, OUTPUT);
  lcd.begin(16,2);
  lcd.setCursor(0,0);
  Serial.begin (9600);
}

long readDistance(int triggerPin, int echoPin)
{
  pinMode(triggerPin, OUTPUT);
  digitalWrite(triggerPin, LOW);
  delayMicroseconds(2);
  digitalWrite(triggerPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(triggerPin, LOW);
  pinMode(echoPin, INPUT);
  return pulseIn(echoPin, HIGH);
}

void loop()
{
  float d1 = 0.01723 * readDistance(t1, t1);
  float d2 = 0.01723 * readDistance(t2, t2);
  float d3 = 0.01723 * readDistance(t3, t3);
  float d4 = 0.01723 * readDistance(t4, t4);

  Serial.println("d1 = " + String(d1) + "cm");
  Serial.println("d2 = " + String(d2) + "cm");
  Serial.println("d3 = " + String(d3) + "cm");
  Serial.println("d4 = " + String(d4) + "cm");

  if(d4 > 200) {
    digitalWrite(Green, HIGH);
    digitalWrite(buzzer, LOW);
    digitalWrite(Yellow, LOW);
    digitalWrite(Red, LOW);
  }
  if(d4 <= 200 && d4 >100) {
    digitalWrite(Green, LOW);
    digitalWrite(buzzer, LOW);
    digitalWrite(Yellow, HIGH);
    digitalWrite(Red, LOW);
  }
  if(d4 < 100) {
    digitalWrite(Green, LOW);
    digitalWrite(buzzer, HIGH);
    digitalWrite(Yellow, LOW);
    digitalWrite(Red, HIGH);
    //tone(buzzer, 100, 100);
  }
}

```

```

if (d1>100 & d2>100 & d3>100){
    lcd.setCursor(0,0);
    lcd.print("3 Slots Free");
    lcd.setCursor(0,1);
    lcd.print("Slot 1 2 3 Free");
    delay(500);
}
else if((d1>100 & d2>100)|(d2>100 & d3>100)|(d3>100 & d1>100))
{
    lcd.setCursor(0,0);
    lcd.print("2 Slots Free");
    lcd.setCursor(0,1);
    if(d1>100 & d2>100)
        lcd.print("Slot 1 & 2 Free");
    else if(d1>100 & d3>100)
        lcd.print("Slot 1 & 3 Free");
    else
        lcd.print("Slot 2 & 3 Free");
    delay(500);
}
else if(d1<100 & d2<100 & d3<100)
{
    lcd.setCursor(0,0);
    lcd.print("No Slot Free");
    lcd.setCursor(0,1);
    lcd.print("Parking Full");
    delay(500);
}
else if((d1<100 & d2<100)|(d2<100 & d3<100)|(d3<100 & d1<100))
{
    lcd.setCursor(0,0);
    lcd.print("1 Slot Free");
    lcd.setCursor(0,1);
    if(d1>100)
        lcd.print("Slot 1 is Free");
    else if (d2>100)
        lcd.print("Slot 2 is Free");
    else
        lcd.print("Slot 3 is Free");
    delay(500);
}

delay(100);
}

```

Code Algorithm

Libraries and Pin Definitions:

- The code includes the `LiquidCrystal.h` library for controlling the LCD display.

- Pins are defined for LCD connections, traffic light LEDs (Green, Yellow, Red), buzzer, trigger and echo pins for ultrasonic sensors (t1-t4).
- A `distanceThreshold` variable sets the minimum distance for a spot to be considered occupied.

Setup Function:

- Sets all defined pins as outputs (for controlling LEDs and buzzer) except echo pins (used for receiving sensor data).
- Initializes the LCD display with 16 columns and 2 rows.
- Starts serial communication for potential data output.

readDistance Function:

- This function takes trigger and echo pin numbers as arguments.
- It sends a short ultrasonic pulse and measures the time it takes for the echo to return.
- The function converts the travel time to distance using a conversion factor (0.01723 cm/microsecond).

Loop Function:

- This function continuously reads distances from all four ultrasonic sensors (d1-d4) using the `readDistance` function.
- Prints the measured distances to the serial monitor for debugging purposes (commented out in your code).

Traffic Light Logic:

- Based on the distance in front of sensor 4 (d4):
 - Green light turns on if the distance is greater than 200 cm (indicating a free spot).
 - Yellow light turns on if the distance is between 100 cm and 200 cm (caution zone).
 - Red light and buzzer turn on if the distance is less than 100 cm (occupied spot).

LCD Display Logic:

- The code checks the state of all three ultrasonic sensors (d1-d3).
- Based on the sensor readings, it displays different messages on the LCD:
 - "3 Slots Free" and lists all slots as free if all sensors detect no obstacle.
 - "2 Slots Free" with details of which two slots are free based on sensor readings.
 - "1 Slot Free" with details of which slot is free based on sensor readings.
 - "No Slot Free" and "Parking Full" if all sensors detect obstacles.

Delay:

- A short delay (100 milliseconds) is added at the end of the loop to prevent overwhelming the system.

Overall, this code demonstrates a basic implementation of a smart parking system using ultrasonic sensors. It can detect car presence in parking spots and provide visual (traffic lights, LCD) and audible (buzzer) feedback.

Conclusion

Debouncing and IMU sensor can be integrated and many more advancements is possible. Timer operation can also be implemented for calculation of the cost for parking in the parking areas of malls ,marriage halls etc.

References

- [1] <https://www.sabelectronic.com/2020/06/arduino-uno-pins.html>
- [2] Data sheet of handson for HC SR04
- [3] Data sheet for PING by Parallax
- [4] [piezoelectronic buzzer ps en.pdf \(tdk.com\)](#)
- [5] [LCD 16x2: Pin Configuration, Features and Its Working \(elprocus.com\)](#)